ISEAS YUSOF ISHAK INSTITUTE

ISSUE: 2025 No. 43
ISSN 2335-6677

PERSPECTIVE

RESEARCHERS AT ISEAS – YUSOF ISHAK INSTITUTE ANALYSE CURRENT EVENTS

Singapore | 12 June 2025

Data Centres, Energy Demand and Sustainability: Can Malaysia Strike the Right Balance?

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Malaysia has ambitions to become a digital powerhouse. The country has placed data centres at the forefront of its economic strategy to achieve this goal, attracting RM184.7 billion (US$43.6 billion) in investments in data-centre-related projects between 2021 and 2024. This photo on 18 November 2024 shows the topping out ceremony for the Infinaxis Data Centre in Cyberjaya, Malaysia.

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ISSUE: 2025 No. 43
ISSN 2335-6677

EXECUTIVE SUMMARY

- Malaysia's data centre sector has attracted RM184.7 billion (US$43.6 billion) in investments in related projects between 2021 and 2024. Data centres are mainly situated in Johor and Klang Valley. Data centre energy consumption in Malaysia is projected to surge to over 5,000 MW by 2035, i.e., 40 per cent of Peninsular Malaysia's present power capacity, or 11.1 per cent of Malysia's projected power capacity in 2035.

- At present, 38 projects have secured Electricity Supply Agreements with Malaysia's primary electricity utility company Tenaga Nasional Berhad (TNB). The actual load utilisation as of December 2024 was 405 MW (3 per cent of TNB's total supply) while the total maximum demand secured is 5.9 GW (43 per cent of TNB's total supply), a hazardously high figure.

- A potential implication of this overexpansion is the rise in electricity prices for residential ratepayers, as seen in more mature data centre markets.

- To address the increase in energy demand, Malaysia's current approach relies on incentivising optimisation of Power Usage Effectiveness (PUE) and related metrics, and shifting to renewable energy. However, focusing on these metrics runs the risk of greenwashing rather than achieving sustainable growth.

- Moving forward, Malaysia needs to consider how it can make the data centre industry pay for its full cost of service for electricity without adding costs for residential ratepayers. Malaysia should go beyond an incentive-based approach and develop proactive, enforceable guidelines that balance growth and sustainability objectives.

INTRODUCTION

Malaysia has ambitions to become a digital powerhouse. To achieve this goal, the country has placed data centres at the forefront of its economic strategy, attracting RM184.7 billion (US$43.6 billion) in investments in data-centre-related projects between 2021 and 2024.[1] Data centres are physical facilities that house computer systems and related infrastructure, such as servers and storage, used to store and process data. They form the foundations of the country's computational power, a core dependency in building large-scale Artificial Intelligence (AI).[2]

Since last year, Prime Minister Anwar Ibrahim has explicitly called for a more selective approach towards foreign direct investments, accepting data centre investments primarily in the field of AI and data technology.[3] He distinguished these AI data centres from "data centres of the past that used to depend on cheap energy and water to move and cool the operating systems"—the latter he considered to be "no longer sufficient". Yet, these AI data centres are even more power intensive. A typical AI data centre, according to the International Energy Agency (IEA), uses as much power as 100,000 households at present, while the largest centres currently being constructed will consume 20 times that amount.[4] The computational power needed for sustaining AI's growth is doubling roughly every 100 days.[5] It is thus of no surprise that data centre energy consumption in Malaysia is projected to surge to over 5,000 MW by 2035, i.e., 40 per cent of Peninsular Malaysia's present power capacity, or 11.1 per cent of Malaysia's projected power capacity in 2035.[6]

Globally, much work has been done to investigate the environmental impact of energy- and water-intensive data centres, whose servers are kept on and cold 24h/7 all year round.[7] Yet, conversations with Malaysia's policymakers reveal that Malaysia's current focus is to ensure the benefits of AI are being realised, with no landing as of yet on the narrative surrounding how to deal with the corresponding increase in energy demand.

It is thus time to bring the conversation about energy requirements into policy circles in Malaysia. After all, data centre investments have been a key part of the Anwar administration's narrative of spurring economic growth. This Perspective examines this issue in four sections. Following this introduction, the second section will examine the implications of data centre growth and increasing energy demand. The third section analyses Malaysia's current dual-pronged approach to data centre development. Finally, the paper concludes with policy recommendations.

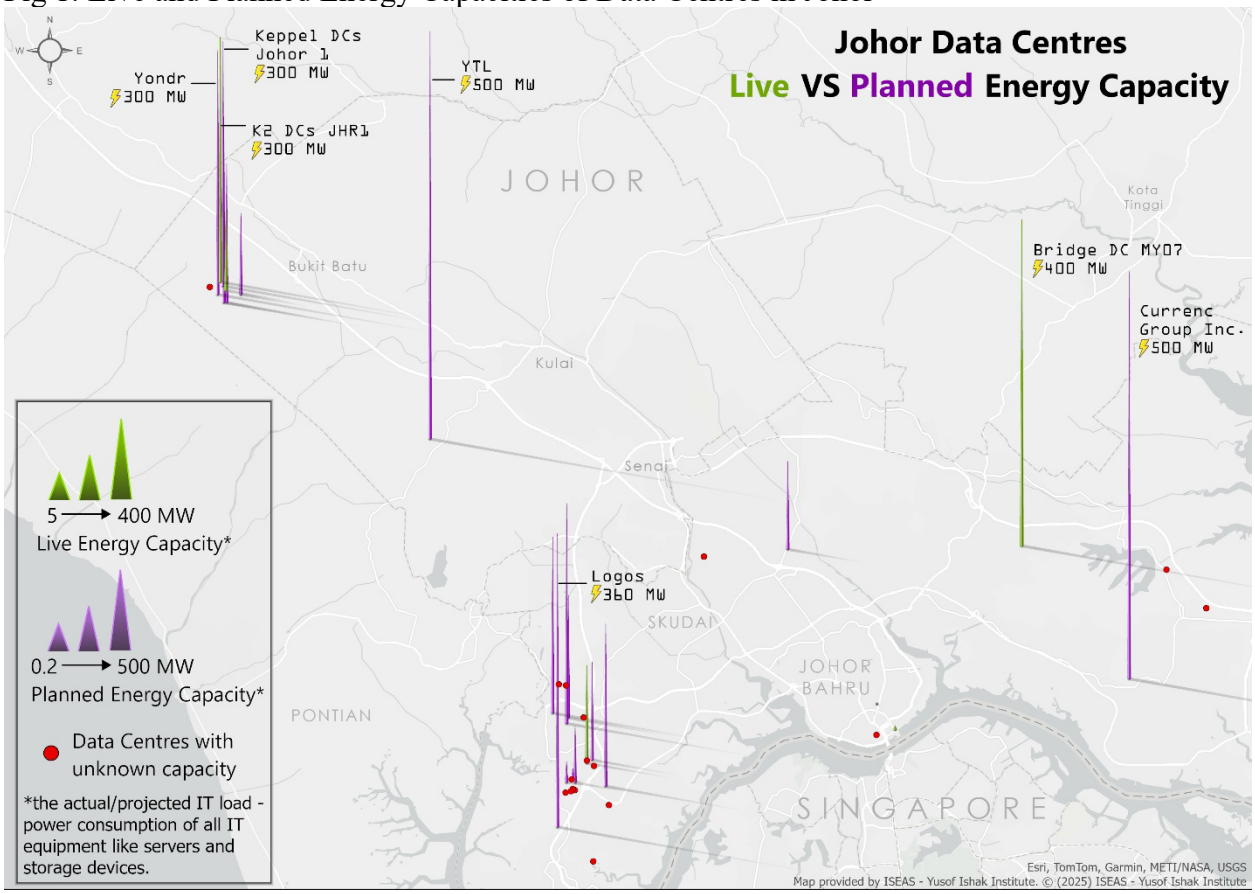
DATA CENTRE GROWTH AND INCREASING ENERGY DEMAND

Data centres are not a new development in Malaysia. Private data centres have existed for decades, constructed and located either within an organisation's premises or at an external location for exclusive use. These data centres are entirely managed by the organisation. What is different about the current influx of investments is the scale of data centres; standalone ones cover a few hundred square metres of data centre halls. These data centres can be broadly categorised as hyperscale or co-location. Hyperscale data centres generally contain at least 5,000 servers and occupy at least 10,000 square feet of physical space, typically consuming between 100 megawatts (MWs) and 300 MWs.[8] Co-location, on the other hand, is a situation wherein one company owns a hyperscale data centre and rents out its facilities, servers, bandwidth and/or space to interested businesses that presumably do not have their own data centre facilities.[9] Examples of hyperscalers include Amazon Web Services' planned facility in Cyberjaya and Microsoft's planned facilities in the Malaysia West cloud region.[10] Other

hyperscale investment projects in Malaysia's Johor—the fastest growing market within Southeast Asia with over 1.6 GW of Total Supply, as of 2024—include two of AirTrunk's planned data centres (150 MW and another scalable to over 270 MW) and Yondr Group's planned 300 MW data centre development in Johor's Sedenak Tech Park (see fig. 1).[11]

Malaysia's data centres are concentrated in two main areas—the southern tip of Johor, at the southern end of the Malaysian Peninsular bordering Singapore, and Klang Valley, the urban expanse from Kuala Lumpur to Port Klang, including Cyberjaya. As of 2024, 1.8 gigawatts (GW) of Early-Stage Supply sits between Kuala Lumpur and Johor Bahru in a 55:45 split. According to Johor state's data centre development coordination committee, the state's data centre capacity has grown from 10 megawatts (MW) at the start of 2021 to around 1.3 gigawatts (GW) by November 2024, and that figure is expected to rise to 2.7 GW by 2027.[12]

Fig 1: Live and Planned Energy Capacities of Data Centres in Johor



Credits: Ronel Giam/ISEAS – Yusof Ishak Institute
Source: Author's compilation[13]

There are three main drivers for this recent growth in data centres. Firstly, demand-side factors include the growth of cloud and AI, along with the globally increasing demand for storage capacity and data processing to perform critical elements in daily tasks such as social networks, e-commerce and data storage.[14] Secondly, supply-side factors include the availability of resources such as power and water, fibre connectivity, its potential as a Johor Cable Landing Station (CLS) for the submarine cable telecommunications network, and land availability.[15] Thirdly, a key external development that has accelerated the growth is Singapore's imposition of a three-year moratorium on data centres in 2019, subsequently replacing it with stricter conditions on the data centre usage of electricity, water and land. Malaysia on the other hand

has streamlined power approvals, with TNB launching the Green Lane Pathway initiative in 2023 which fast-tracks supply offerings for electricity. With this initiative, data centres are connected three times faster than the normal delivery time, reducing the implementation period of 36 to 48 months, to 12 months.[16] The government has approved a total of 21 data centre projects under the Digital Ecosystem Acceleration Scheme, with a total investment of RM 113.8 billion (US$ 26.8 billion) since 2021.[17] Johor, in particular, is considered an ideal location due to connectivity to Singapore. As seen in Fig. 1, many data centres are concentrated near the border with Singapore in Nusajaya Tech Park.

As seen in Table 1 below, while Klang Valley has the highest number of data centres, Johor leads the nation in terms of IT capacity, accounting for close to 80 per cent of the total live IT capacity in the country as of 2024.[18] These figures are rapidly rising—based on market research firm DCByte's database as of May 2025, Johor has 47 "live + pipeline" (up from 2024's 40) projects, while Klang Valley has 68 (up from 65), though data for the corresponding IT capacities and breakdown between live and pipeline are not available.

Table 1: Live and Pipeline Energy Capacity of Data Centres in Johor and Klang Valley (2024)

		Johor	**Klang Valley**	**Johor + Klang Valley**
Number of data centres	**live**	12	37	49
	pipeline	28	28	56
	live + pipeline	40	65	105
IT capacity/ electricity demand (MW)	**live**	396.9	107	503.9
	pipeline	898.7	378.5	1277.2
	live + pipeline	1295.6	485.5	1781.1

Source: Knight Frank, 2024

TNB's report further indicated a cumulative 38 projects secured Electricity Supply Agreements (ESA) with a total maximum demand of 5.9 GW as of December 2024. In 2024 alone, 9 data centre projects were completed with a total maximum demand of 1.3 GW.[19] The actual load utilisation reported was 405 MW for December 2024. Based on TNB's total contracted capacity of 13.76 GW,[20] the actual load utilisation is about 3 per cent of TNB's total supply; the maximum demand based on existing projects alone however stands at close to 43 per cent. This figure is hazardously high. While utilisation is rolled out in phases, it is worth questioning if Malaysia has overcommitted to attracting and developing data centres without sufficient (growth in) energy capacity to support this expansion.

This concern is more pronounced when considering that as AI usage shifts from training to inference (application/use) stage, and as data centres shift from production to use stage, energy consumption is also projected to increase. The inference phase accounts for approximately 60–70 per cent of overall machine learning-related energy use and 20–40 per cent in the training phase.[21] Greenhouse Gas (GHG) emissions of data centres during the use phase in 2020 have

been estimated at 95 MtCO2e, which is three times greater than the GHG emissions of the production stage.[22]

Potential Implications: Rise in Electricity Prices

In more mature data centre markets, the influx of data centre investments has contributed to higher electricity prices for residents and small businesses. Malaysia should take a leaf from these case studies as it attempts to strike a balance between growth and sustainability. In Georgia, US, the fastest growing market for data centres in the US, Georgia Power reported that 80 per cent of its projected increased energy demands by 8,200 MW by 2030 is tied to prospective data centres opening in the state.[23] To cope with the increased demand for electricity, base electricity rates have been driven up by excess fuel expenses and the construction of new nuclear power generators and miles of transmission lines.

Since 2023, the average Georgia Power residential customer is paying US$43 more per month following increases to base electric rates.[24] Like Malaysia, Georgia was an attractive market for data centres, given the relatively low electricity prices, with industrial electricity rates approximately 42 per cent below the US national average. Generous tax breaks were also promised, with at least US$163 million waived in local state and sales tax collections each year since 2022.[25]

To respond to these challenges, Senate Bill 34 was filed to protect residential and commercial ratepayers from facing higher utility bills because of large-scale utility company investments to meet the energy demands of artificial intelligence.[26] The bill seeks to prohibit certain costs incurred by an electric utility as a result of providing electric services to commercial data centres.[27] A data centre rule authorised Georgia Power to enact contractual agreements with companies that will use 100 MW of power, allowing the utility company to "appropriately assign" to data centres the costs of providing them electric service.[28] The rule also gives Georgia Power the ability to require data centre companies to put up front-end collateral for energy costs over the lifetime of the contract, which allows Georgia Power to keep the remaining money owed if the company abandons the project prior to the contract expiring.

Similar discussions about the rise of data centres contributing to higher electricity rates have emerged in Malaysia. In December 2024, TNB which operates the power grid in Peninsular Malaysia announced a proposed adjustment of the base electricity tariff to 45.62 sen/kWh in July 2025 from 39.95 sen/kWh, a 14.2 per cent increase.[29] Although Prime Minister Anwar has denied an increase in tariffs of this magnitude, he confirmed a "slight increase" in electricity tariffs in Peninsular Malaysia.[30] While the primary reason cited for the increase in tariffs has been higher fuel costs assumption, costs to operate, maintain and upgrade the national grid are also contributing factors. TNB disclosed that it would raise its grid and distribution network capital expenditure to RM26.5 billion in RP4 (2025-2027), up from RM20.55 billion in RP3 (2022-2024). It also announced increasing operating expenditure to RM20.782 billion from RM17.96 billion in RP3. The higher expenditures may be partly explained by the need to maintain data centres which require electricity 24h/7.[31]

Several criticisms have been raised, including the passing on of increased prices to consumers.[32] It is crucial for Malaysia to consider more seriously if the data centre industry should be paying a premium for electricity, rather than adding costs for residential ratepayers. While there have been talks of global technology companies being expected to pay a premium

for energy and water access to support their data centres in Malaysia, there have not been any policy interventions of this nature as of yet.[33]

CURRENT POLICY APPROACH

To cope with the increase in demand for energy, Malaysia's current approach is one of mitigating the environmental impacts of data centre operations.[34] Underpinning this approach is the belief that the dual pronged approach of optimising Power Usage Effectiveness (PUE) and related metrics, and shifting to renewable energy, is sufficient for the sector to grow sustainably. The latest policy update—a press release by the National Investment Council in December 2024—recommended the PUE for hyper-scale data centres (with a capacity of more than 21MW) to be 1.4 or lower, for consideration for tax incentives.[35] This means that the total energy consumed by the data centre facility should be no more than 1.4 times of the energy used by the IT equipment (as mapped out in fig. 1). For example, if Bridge DC's planned IT capacity is 400 MW, its data centre facility consumption should ideally be no more than 560 MW. This guideline builds on the Johor State Planning and MIDA guidelines, which are summarised in the tables in Appendix 1.

While these guidelines serve as an incentive for optimising PUE, they are not enforceable. The recommended PUE of 1.4 and below is for firms to be considered for tax incentives. As seen from the bolded words in the tables (see Appendix 1), many points are framed as recommendations rather than enforceable regulations. These include the guideline about using renewable and energy-efficient technologies. These guidelines also lack specificity about the estimated reduction in carbon emissions required through such energy efficient technologies, be it energy-efficient hardware, AI-optimised cooling, or smarter data centre design and operations.[36]

In addition, while there has been recognition of the detrimental downstream impacts of data centres, policymakers developing AI ecosystems struggled to comment on a strategy to address the energy needs. In an interview, the National Artificial Intelligence Office (NAIO) CEO Sam Majid stated that, "we are still in the midst of making sure that the benefits of AI are being realised. Although some people do appreciate that there is the associated energy cost that comes with it, most of the ecosystem is still banking on the benefits, the features and the outlook of what AI can be […]. On a regular basis, the key agencies and partners of the energy or resource component interface with NAIO, but we haven't come to a landing yet on what the next narrative would be on those kinds of conversations."[37] Conversations with regional offices like the Iskandar Regional Development Authority (IRDA) reveal similar findings. Despite having made progress towards reducing carbon emissions through implementing their many blueprints towards a Low Carbon Society for Iskandar Malaysia over the years, a representative shared that they are only just starting to research the carbon emissions produced by data centre developments.

Risks of greenwashing

It is questionable if present guidelines can truly be considered "sustainable". Using this term in fact runs the risk of greenwashing. In more mature data centre markets, experts have warned that the total environmental impact could remain the same or increase, just with the perception of a "greener" data centre.[38]

Thus far, AirTrunk has signed a Virtual Power Purchase Agreement of 30 MW, with developer ib vogt, under Malaysia's Corporate Green Power Programme.[39] However, as in the case of Power purchase agreements (PPAs) which have been dominated by big tech (Amazon, Google, Meta and Microsoft were the top 4 purchases of renewable energy in 2022), matching 100 per cent of annual demand with renewable energy purchases or certificates does not mean that data centres are powered exclusively by renewable sources. Wind and solar power may not always meet a data centre's energy demand, and renewable energy may have been purchased from projects in a different location to the demand.[40]

While Malaysia sought to facilitate the transition to renewables by opening its national grid in September 2024 to enable corporate users to directly engage with renewable energy companies, the capacity of solar panels to power data centre has been minimal thus far.[41] AirTrunk's 150 MW hyperscale data centre JHB1 in Johor, equipped with a solar-ready roof, has a capacity of 1 MW, providing a mere 2 per cent of its current phase's energy needs.[42] For the most part, data centres in Malaysia draw power from "dirty" electricity grids—in 2023, total renewable energy (RE) made up only 8.3 per cent of the total electricity TNB produced, while the majority was coal and gas.[43]

Industry players have also shared that the intermittent nature of solar energy (at least in the absence of a well-developed battery storage systems) does not make it an ideal energy source for data centres, given the need to keep the data centres running 24h/7. With limited solar generation capacity, data centres in Malaysia often rely on back-up diesel generators. Solar panels also require land, making it less practical as an option. Ten acres of land is typically required to generate 1 MW of power.[44] Based on the current total maximum demand of 5.9 GW based on the 38 data centre projects that have signed ESAs, that would require 59,000 acres of solar panels. That makes up over 60 per cent of Johor's landmass of 96,680 acres. While renewable diesel is an available option, there are currently no regulations that encourage this transition.

Industry players have also expressed concerns that the costs of renewables are higher than sticking to conventional sources of the grid. This difference may be explained by the wheeling charges incurred, or system access charge (SAC) as per the Energy Commission's guidelines. Higher SAC is imposed on Independent Power Producers (IPPs) for producers that generate electricity intermittently (at 45 sen (10 US cents)/kwH) compared to those capable of supplying firm output (at 25 sen/kwH).[45] While the data centre sector has been identified as a potential area for green transition, incentives for increased reliance on solar energy is not clear, given that solar is an intermittent renewable energy source with higher SAC charges. At present, Bridge Data Centre has signed a bilateral Energy Supply Contract for Green Energy Supply of 400 MW via the Corporate Renewable Energy Supply Scheme (CRESS), a system that offers open grid access where a third party can supply or buy electricity through the grid network system with a set SAC.[46] These challenges, as observers have pointed out, may be mitigated by IPPs investing in battery storage facilities that allow IPPs to provide a stable supply, though the additional cost of energy storage is likely to be passed on to customers through higher tariffs.[47]

CONCLUSION

Malaysia's ambitions to be a digital powerhouse demand urgent consideration of rapidly increasing energy demands. Based on current figures, Malaysia appears to have overcommitted to data centre expansion, especially in Johor, without sufficiently robust sustainability

guidelines in place. The current approach towards sustainable data centre development relies on optimising PUE and related metrics, and recommends transitioning to renewables. These run the risk of greenwashing and without more robust guidelines in place, the overexpansion has led to early signs of adverse downstream impacts on local communities and the environment.[48]

Moving forward, Malaysia first needs to consider how it can make the data centre industry pay for its full cost of service for electricity without adding costs for residential ratepayers. This may be achieved via differential pricing for large users of energy over a certain threshold (for example, based on the current policy definition of hyper-scale data centres as having a capacity of more than 21MW). Announcements of the impending increased electricity tariffs have already led to concerns that Malaysia would be less attractive to investors. However, the flipside would be that investments are attracted by providing electricity at a discount at the expense of local consumers. Secondly, beyond optimising for PUE and related metrics, a lifecycle assessment that maps out the end-to-end stages of data centre energy consumption offers a more holistic approach.[49] Key areas of importance are energy consumption during the in-use phase (related to energy efficiency), the embodied impact of materials in the data centre (including IT equipment and M&E services) and the impact of the source energy (e.g. renewable or fossil fuel based) for all phases, i.e. not just in-use but also during manufacture and other stages.[50] Malaysia needs to move beyond an incentive-based approach and develop more proactive, enforceable guidelines to strike a better balance between growth and sustainability.

ISSUE: 2025 No. 43
ISSN 2335-6677

APPENDIX

Appendix Table 1: Johor state planning guidelines for data centres (April 2024)

Electrical energy requirements	Water supply requirements*
• Ensuring the presence of an electrical substation with a capacity of 33 kV for small data centres and 132 kV to 275 kV for large-scale data centres • Data centre development is **encouraged** within a 100-metre radius of transmission lines or Main Intake Substations (PMU) • All data centres under development must strictly adhere to industry best practices for Power Usage Effectiveness (PUE) to optimise energy efficiency and reduce operational costs • The provision of either a conventional generator set or green technology is essential and will be subject to the development capacity of the data centre being constructed • Data centres **should** also incorporate the use of renewable and energy-saving technologies • Any requirements related to electricity supply must be reviewed and approved by Tenaga Nasional Berhad (TNB) before the data centre development proceeds.	• It is imperative for a data centre development to meet the minimum daily water supply requirement. • Provision of water storage tanks is determined by the water consumption of the data centre. • Data centres **should** utilise renewable and energy-efficient technologies, such as direct expansion systems and eco-chiller water systems for cooling, as well as Rainwater Harvesting Systems (RWHS) • Review and approval by relevant authorities such as authorities, including Ranhill SAJ Sdn. Bhd., the National Water Services Commission (SPAN), the Johor Water Regulatory Body (BAKAJ) necessitated. • Data centre development must incorporate water conservation strategies that align with industry-recognised standards for water usage effectiveness, as prescribed by the prevailing norms and best practices

** In addition to these guidelines, SPAN's guidelines on water use are in the formulations and will be concluded by mid-2025, according to former SPAN Chairman Charles Santiago.*

Appendix Table 2: MIDA – guideline for sustainable development of data centre (for tax incentives under the Digital Ecosystem Acceleration (DESAC)[51] scheme)[52]

Power Usage Effectiveness (PUE)	Carbon Usage Effectiveness (CUE)	Water Usage Effectiveness (WUE)
The organisation (owner/ developer/ operator) of the data centre **should** accelerate energy efficiency measures with best-in class technologies at the hardware and software level, and the design PUE limit shall be based on different category of data centre	The organisation may purchase renewable energy resources for its operation or produce or co-producing (renewable) electricity on-site or generating CO2 in other manners (e.g., maintenance operation of diesel generators)	The organisation **should** avoid water stress areas in locating the new data centre by considering areas with water stress index (WSI) of less than 0.8 (only applicable for Peninsula Malaysia) The organisation **should** deploy water efficiency management

		practices and innovation to accelerate the efficient use of water by the data centre. The organisation may utilise the reclaimed or reused water for their operation. The recommended design WUE is 2.2 m3/MWh or lower, and the organisation shall continuously monitor and disclose the actual WUE in their annual sustainability report and work to improve the WUE of 2.0m3/MWh or lower from the baseline design WUE value over the next 10 years.
The organisation shall declare the calculated design PUE value according to international standard ISO/IEC 30134-2.	The organisation shall declare the calculated design CUE value according to international standard IS0/IEC 30134-8.	The organisation shall declare the calculated design WUE value according to international standard ISO/IEC 30134-9.
PUE of 1.4 or lower for hyper-scale data centres (with a capacity of more than 21MW).	Good practices in controlling carbon emissions CUE based on the Grid Emission Factor value set by the Energy Commission to measure carbon emissions by data centres: 0.758 GgCO2 /GWh for Peninsular Malaysia, 0.425 GgCO2 /GWh for Sabah and 0.198 GgCO2 /GWh for Sarawak.	WUE of 2.2 m3/MWh or lower.

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